

# FFCCS External Data Request Form

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## Fire Fighter Cancer Cohort Study (FFCCS) - External Researchers Data Request Procedure and Application Created on: 05/22/2023 Last Updated on: 07/15/2024

FFCCS data types and request overview: The FFCCS collects and integrates survey, biological, and environmental sampling data from a nation-wide cohort of firefighters. Data and specimens are stored and managed by the University of Arizona Center for Firefighter Health Collaborative Research. Only data without identifiers (e.g. enrollee names, contact information or fire department) will be released. In some instances, potentially identifiable data (e.g., sex, race) may not be released if individuals may be indirectly identified because of small cell size (e.g., < 10) numbers. Any contact with FFCCS participants (outside of an existing funded/approved research project under the direction of the Principal Investigator for that project) must be approved by FFCCS leadership and be conducted following either a protocol that includes an existing FFCCS researcher or a protocol with the explicit support of the FFCCS leadership team.

The FFCCS Co-lead team and Director will use the following review guidelines according to the type(s) of data requested.

### Data type(s)

[Only data without identifiers can be released]

FFCCS reviews and approvals needed

#### 1. Survey data only

- Data Coordinating Committee (DCC) Co-Chairs
- Option for FFCCS Co-leads by email (including OPB Steering Committee)
- FFCCS Director

#### 2. Survey data linked to either (or both):

- existing test results from biospecimen (e.g. PFAS, PAHs)
- Results of previous environmental sampling (e.g. PAHs)
- DCC and/or Biomarker Analysis Center (BAC) and Exposure Assessment Center (EAC) Co-Chairs
- FFCCS Co-leads by email with required response
- OPB (oversight and planning board) Steering Committee (including OPB co-chairs)
- FFCCS Director

#### 3. Biospecimens samples for analysis with linked survey or environmental data

- Review (by email) of the FFCCS Co-leads chairs (including OPB Steering Committee)

Expectations: It is an expectation of the FFCCS that results are shared with participants. This need not be individual results but can include a lay-person summarization to be shared with participants prior to any public presentation.

## I. Data Request Procedure:

Data request application process:

- All data requests from external researchers must be received through the online and form and must be completed and submitted through the online data request portal at [you are currently in the data request portal].
- For the purposes of this application, an external researcher is defined as anyone not listed as an investigator on any of the FFCCS-IRB approved projects from which data are being requested.
- External researchers are strongly encouraged to discuss their data requests with one of the FFCCS Co-leads prior to submitting their data request.
- The application will be reviewed by the University of Arizona Data team and the requestor contacted for any outstanding questions. This is a pre-review to ensure application completion and does not infer approval.
- Once received, the application will be reviewed by the data application subcommittee of the FFCCS Co-lead team.
- The subcommittee will include at least one member from the Data Coordinating Committee (DCC), Exposure Assessment Core (EAC), Biomarker Assessment Core, and the Oversight and Planning Board (OPB).
- The subcommittee will make a recommendation to the FFCCS Co-leads who will make a final determination and report all data request decisions to the larger FFCCS meeting.
- The DCC will determine whether a data use agreement is needed and between which institutions.
- All approved data requests will be logged on the online spread sheet housed at: [www.ffccs.org/LINK](http://www.ffccs.org/LINK).

## II. FFCCS publication process and co-authorship requirements:

- Any products resulting from this data request must be submitted to the FFCCS publication committee.
- Publications using FFCCS data are pre-reviewed (an abstract and outline of the paper) by the FFCCS publication committee. FFCCS coauthors should be involved in the drafting or review of the manuscript before it is submitted to the FFCCS publication committee.
- Conference abstracts must be submitted at least two weeks in advance to be reviewed and approved by the FFCCS Publication Committee. Proposal should be submitted through the FFCCS website portal [[www.ffccs.org/](http://www.ffccs.org/) ...]
- FFCCS publication process and co-authorship requirements:
- Any authored products resulting from this data request must minimally include the PI of the respective FFCCS project(s) from which data are shared.
- Funding sources for each project with data included in the product must be acknowledged.

### Data Request Form

#### Request/requestor Information

Date of request 02-21-2025

#### Requesting investigator information (or Mentor of Trainee):

Investigator First and Last Name Janine LaSalle

Investigator Credentials (e.g. MD, PhD) PhD

Investigator Academic or Institutional Title (e.g., Assistant Professor) Professor

Investigator Institution University of California Davis  
(requester institution)

|                                 |                                               |
|---------------------------------|-----------------------------------------------|
| Investigator Department Address | Medical Microbiology and Immunology           |
| Investigator Contact Phone      | (530) 754-7598                                |
| Investigator Contact Email      | jmlasalle@ucdavis.edu<br>(email of requestor) |

**Mentee/Trainee Information**

|                                                                            |                                                           |
|----------------------------------------------------------------------------|-----------------------------------------------------------|
| Mentee/Trainee First and Last Name                                         | Logan Williams                                            |
| Mentee/Trainee Credentials (e.g. MD, PhD)                                  |                                                           |
| Mentee/Trainee Academic or Institutional Title (e.g., Assistant Professor) | Graduate student researcher                               |
| Mentee/Trainee Institution                                                 | University of California Davis<br>(requester institution) |
| Mentee/Trainee Department Address                                          | Medical Microbiology and Immunology                       |
| Mentee/Trainee Contact Phone                                               |                                                           |
| Mentee/Trainee Contact Email                                               | lwilli@ucdavis.edu<br>(email of requestor)                |

**Proposed Project**

|                                 |                                                                                                     |
|---------------------------------|-----------------------------------------------------------------------------------------------------|
| Title of request/project        | Examining Longitudinal Changes in DNA Methylation in Firefighters Exposed to Products of Combustion |
| Date needed by (if applicable): | 04-01-2025<br>(note the team will adhere to this date as possible)                                  |

Background: up to 750 words with references (not included in word count)

Brief description of study/data analysis being completed with requested data

Aim 1. Identify DNAm regions sensitive to combustion products' exposure and dose, prioritize DNAm regions for inclusion in FECS, and test the efficacy of each DNAm locus in FECS for intervention-sensitive changes. (LaSalle lab, UC Davis)

Hypothesis: Firefighter instructors' exposure to carcinogenic agents from products of combustion during training fires will result in quantifiable and dosage-sensitive changes to circulating plasma DNA methylation of gene loci relevant to risk for multiple cancer types and stably maintained after exposure.

Approximately 160 blood plasma samples that were collected pre- and post-exposure from 40 training instructors and 40 recruits in January 2023, plus 40 samples from former instructors in the FFCCS will be sent to Dr. LaSalle's lab for cell free DNA isolation, whole genome bisulfite sequencing, and analysis of differentially methylated regions.

## Funding Sources

**While a funding source is not required, proposals without peer review (competitive external funding source or internal pilot funding) will require a full technical review of the FFCCS co-leads**

Is this project part of a training grant?

☐ Yes  
☒ No

Funding Amount

\$747,000

Funding Years

12/1/24-11/30/27

Is this project part of an academic degree requirement?

☐ Yes  
☒ No

## Data or specimen elements request

Which FFCCS project or projects are included in your data or specimen request, check all that apply

- ☒ FFCCS Framework (2016-2019)
- ☐ Expansion of the Fire Fighter Cancer Cohort Study: Arson Investigators, Instructors, Volunteers, and Wildland-Urban Interface Firefighters (2018-2022)
- ☐ Per- and Polyfluoroalkyl Substances (PFAS): Firefighter Exposures and Toxicity (2019-2023)
- ☐ Wildland Urban Interface Study using Disaster Aid Supplemental Funding (2020-2022)
- ☐ Women Fire Fighters Study: Stress, Cancer Risk and Reproductive Toxicity (2020-2025)
- ☐ Volunteer Firefighter Cancer Assessment and Prevention study Volunteers (CAPS; 2021-2023)
- ☐ Wildland Fire Fighters: Exposure Assessment, Carcinogenic Effects and Risk Management (2021-2026)
- ☐ Electric Battery Fire Sub-study
- ☐ Wildland Urban Interface R01 (2024-2029)
- ☐ North Carolina Firefighters Sub-group
- ☐ High Risk Incident Sub-study
- ☐ Cancer Screening Sub-study
- ☐ ABOR Firefighter Collaborative Research Project
- ☐ Firefighter Sulforaphane Pilot Study
- ☐ Reproductive Outcomes in Male Firefighters Study
- ☐ Per- and Polyfluoroalkyl substances (PFAS) Exposures and COVID-19 in Firefighters (2021-2022)
- ☐ Other: please specify \_\_\_\_\_

Type of data and/or specimens are you are requesting (check all that apply):

- ☒ Survey data (Participant Completed)
- ☒ Biological specimens
- ☐ Biological data (PFAS, epigenetic age)
- ☐ Exposure data (wristband data)
- ☐ Exposure specimens

Subject inclusion criteria for each of the studies selected above, be specific

From LACFD training in January 2023, choose paired blood plasma samples from pre- and post-exposure from 40 training instructors and 40 recruits, matched for gender. Also, choose 40 blood plasma samples from former instructors in the FFCCS.

Subject exclusion criteria for each of the studies selected above, be specific

2 ml of plasma is ideal, but we can work with 1-2 ml. The main exclusion criteria will be participants without pre- and post plasma volumes >1 ml each. We understand the sample are mostly from male participants. We do not plan to exclude females, but if a similar M:F ratio could be used for all three categories (recruits, instructors, former instructors), that would be ideal.

## Survey Data

[Click here to download the current FFCCS Data dictionary for the Baseline Survey](#)

[Attachment: "FireFighterCancerCohortStudyUA\_DataDictionary\_BASELINE\_2025.01.31.xlsx"]

## Baseline Variable Groupings

Choose the variables groups you are interested in requesting.

- ☒ Demographics and General Info
- ☐ AFFF & PFAS Exposure
- ☐ Turnout Gear
- ☐ Drinking Water
- ☐ Exposures outside the Fire Department
- ☒ Healthcare Utilization
- ☐ Sun Exposure and Skin Care
- ☐ Sex Specific
- ☐ Stress Survey
- ☒ Individual (Self) Cancer History
- ☒ Family Cancer History
- ☒ Tobacco and Alcohol Use
- ☐ Building Exposures
- ☐ Ionizing Radiation Exposures
- ☐ Household Pesticide Use
- ☐ Occupational History
- ☐ Military History
- ☐ COVID-19
- ☐ PFAS

[Click here to download the current FFCCS Data dictionary for the Annual Follow-Up Survey](#)

\*Coming summer 2024

## Annual Follow-Up Variable Groups

\*Coming summer 2024

**Biological Specimens**

**Note: Requestors must provide a log of all specimens used and return any unused samples**

Specimen Type Request Volume (mL) Number of aliquots/sample Endpoints

Whole Blood (From EDTA) \_\_\_\_\_  
 Whole blood (from Tempus Tube) \_\_\_\_\_  
 Serum \_\_\_\_\_  
 Plasma (from EDTA Tube) Yes 1.0 mL 2 cell free DNA  
 Plasma (from Na Citrate Tube) \_\_\_\_\_  
 Urine \_\_\_\_\_  
 Other, Please Specify \_\_\_\_\_

Plasma (from EDTA tube)  
 Please provide a description of the methods of testing, include the volume needed for each:

10-15 ng of cell free DNA will be isolated from 2 ml of frozen human plasma using the QIAamp Circulating Nucleic Acids kit.

Please give a detailed description how samples will be handled (sample tracking, storage, etc.). Please note that all shipping costs must be paid for by the requestor.

We will provide a FedEx shipping account, address, and contact information for samples to be shipped on dry ice. At UCD, plasma and DNA samples will be labelled, recorded into FreezerWorks, and stored at -80C.

What is the timeline for completion of testing once sample are received?

2 months for DNA isolation, 6 months for sequencing, 6-12 months for bioinformatic analyses

What is the timeline for returning unused/remaining samples?

2 months

**Biological and exposure data/previously obtained test results**

Analytes or specific test results requested

- ☐ Serum PFAS  
☐ Urinary PAH metabolites  
☐ Wristband results  
☐ AMH testing  
☒ Other, Please specify:

Other analytes or specific test results requested:

cell free DNA for DNA methylation analyses

**Data Storage and Protection Plan**

It is an expectation of the FFCCS that results are shared with participants. This need not be individual results but can include a lay-person summarization.

- ☒ Yes  
☐ No

Will results of the project be shared with participants?

What is the data sharing format?

(e.g., IRB-approved results letter template; sample letters and R code for generating them, are available from the DCC. Note that The University of Arizona will be responsible for notification letters if external researchers do not have access to identifiable information.)

In order to accomplish the goal of impacting firefighters' perceived severity and susceptibility of occupational cancer risk associated with their exposures to products of combustion, we will develop a report back protocol for individual and aggregate urinary PAH and VOC findings and aggregate DNAm findings that achieves our communication goals and that is culturally competent in a way that meets the information needs of firefighter research participants.

Specify how data will be stored (e.g., SQL server database) once it has been shared

Data will be backed up on a secure server at UC Davis.

Outline how data will be protected, including any physical (e.g., lockbox), technical (e.g., data encryption), and/or administrative (e.g., control who has access to data) safeguards, if applicable

Administrative safeguards are in place for all data access.

If the investigator requests and is granted access to identifiable data or biospecimen data, the investigator is subject to the requirements of the Certificate of Confidentiality (CoC) under which the data was collected.

\* FFCCS is responsible for notifying investigators requesting to use biospecimens/data of the CoC requirements.

**Data Analysis Plan**

Provide an overview of the planned data analysis plan with enough detail to justify the requested data items

Please upload the preliminary (blank) tables for your planned analysis

[FILE: JL UCOP FF Research Plan.pdf]

**Data Distribution Plan**

Describe the data distribution plan including conference abstracts and planned manuscript(s)

see uploaded research plan for details, but we expect 1-2 manuscripts and multiple meeting abstracts from the proposed research.

I as the requester attest to follow good clinical practice guidelines as they pertain to data stewardship. Any issues associated with the use of the requested data will be reported to the University of Arizona FFCCS team. I am responsible for safeguarding the requested data and all work products derived from this data for the duration of the project. I will permanently safeguard or securely dispose of all requested data once the project is completed.

